## 28 TONS THE MAXIMUM WEIGHT OF NASMYTH'S STEAM HAMMER

The star Sirius A and its almost invisible companion, the white dwarf star Sirius B, whose existence was deduced in 1844 by Bessel.

for powered flight and, in 1842, they designed a **large, steam-powered, passenger plane**. They received a patent for it in 1843 and launched the Aerial Steam Transit Company, which they advertised as flying to exotic locations such as the pyramids, but the idea was eventually abandoned.

In 1842, Austrian scientist **Christian Doppler** (1803–53) suggested how the frequency of sound and light waves vary as objects approach and move away from an observer (see panel, opposite). This is known as **the Doppler effect**. It is these so-called Doppler shifts that later helped reveal that **the Universe is expanding** (see 1929–1930).

The world's first "computer program" was written in 1843 by British mathematician **Ada Lovelace**, daughter of the poet Lord Byron. Lovelace worked with British inventor Charles Babbage (1791–1871) on his Analytical Engine (see 1834), which would have been the world's first computer had it ever been built. Between 1842 and 1843 she worked on translating an article about the Analytical Engine by Italian mathematician Luigi Menabrea (1809–96). As she worked through the article she added a note that included an encoded algorithm, designed to be processed by a machine. If the machine had been built, this algorithm would have been the **first computer program**.


**ADA LOVELACE** (1815–52)

The daughter of poet Lord Byron, Ada Lovelace was a brilliant mathematician. In 1843, she began to publicize British inventor Charles Babbage's ideas for his Analytical Engine, and wrote what is called the world's first computer program for it. She foresaw that Babbage's machine would have uses far beyond mere calculation.

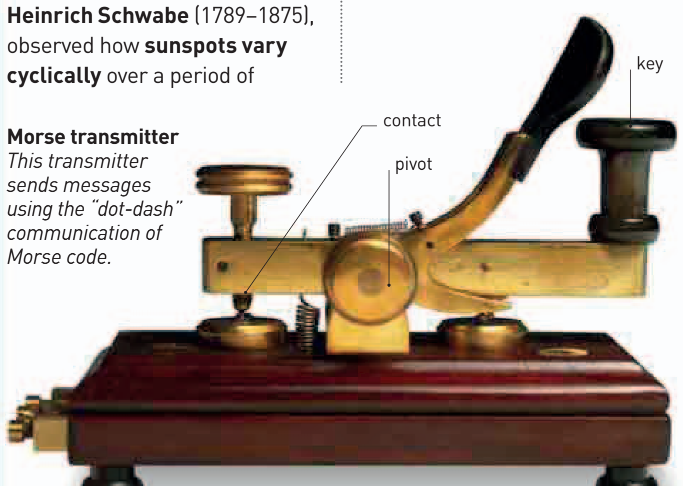
> " CREATURES FAR SURPASSING IN SIZE THE **LARGEST** OF **EXISTING REPTILES…** I WOULD PROPOSE THE NAME OF **DINOSAURIA.** "
>
> **Richard Owen, British Biologist,** in *Report on British Fossil Reptiles*, 1842

## 20 ASTRONOMICAL UNITS THE APPROXIMATE DISTANCE BETWEEN SIRIUS A AND SIRIUS B

**IN 1844, GERMAN ASTRONOMER FRIEDRICH BESSEL** (1784–1846) spotted a wobble in the movement of the stars Sirius and Procyon. Isaac Newton's laws of gravity (see pp.120–21) allow astronomers to calculate the motions of distant stars with such precision that discrepancies can be revealed. Bessel deduced that these **stars have dark companion stars**, now known as Sirius B and Procyon B.

Another German astronomer, **Heinrich Schwabe** (1789–1875), observed how **sunspots vary cyclically** over a period of 10 years—later, this cycle was shown in fact to be 11 years.

Meanwhile, on May 24, American inventor **Samuel Morse** sent the **US's first long-distance telegraph** message down a new line from Washington to Baltimore. It was a Biblical message written in his own Morse code, "What Hath God wrought?"—instant communication had arrived.




**Morse transmitter**
*This transmitter sends messages using the "dot-dash" communication of Morse code.*

**1843** Henson and Stringfellow patent their **steam-powered passenger plane**

**March 25, 1843** Completion of British engineer Marc Isambard Brunel's Thames Tunnel in London, the first bored underwater tunnel

**July 19, 1843** Launch of Brunel's pioneering, iron-hulled steamship *SS Great Britain*

**1843** Ada Lovelace writes the **world's first computer algorithm**

Friedrich Bessel deduces that Sirius and Procyon have **dark companion stars**

**May 24** Samuel Morse sends the **first message using Morse code**

German astronomer Heinrich Schwabe observes cyclic **variation in sunspots**